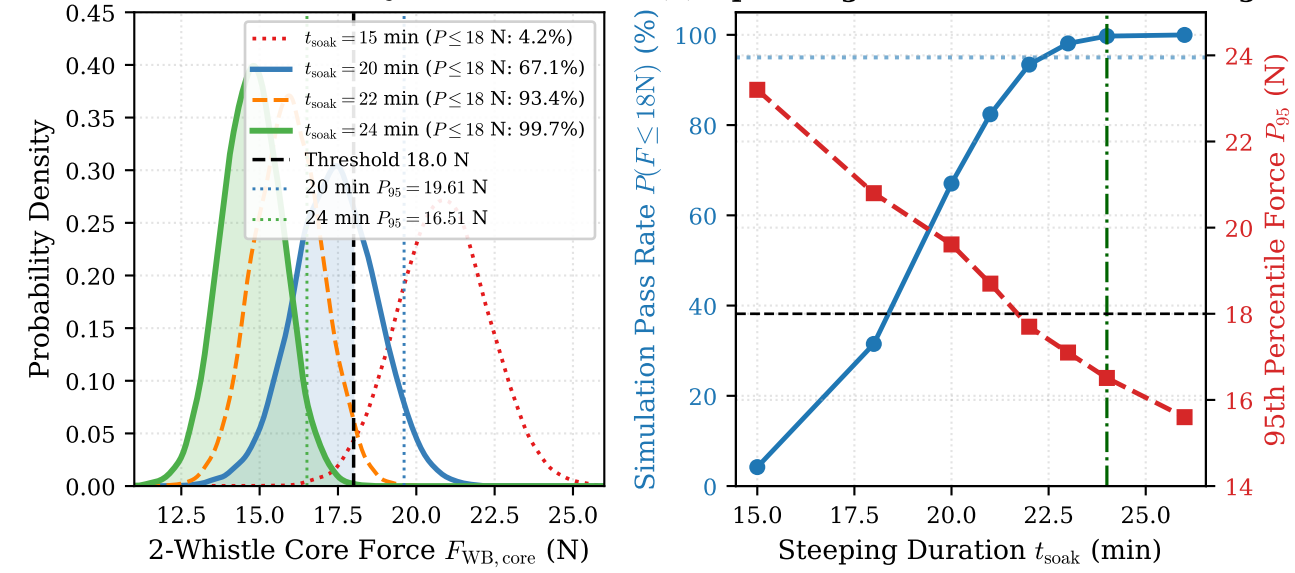
(a) Parameter Uncertainty Distributions ($N = 100$)
(b) Operating Window Robustness Margin
Probability Density
2-Whistle Core Force $F_{\mathrm{WB,core}}$ (N)
$t_{\mathrm{soak}} = 15$ min ($P \leq 18$ N: 4.2%)
$t_{\mathrm{soak}} = 20$ min ($P \leq 18$ N: 67.1%)
$t_{\mathrm{soak}} = 22$ min ($P \leq 18$ N: 93.4%)
$t_{\mathrm{soak}} = 24$ min ($P \leq 18$ N: 99.7%)
Threshold 18.0 N
20 min $P_{95} = 19.61$ N
24 min $P_{95} = 16.51$ N
Simulation Pass Rate $P(F \leq 18\mathrm{N})$ (%)
95th Percentile Force P_{95} (N)
Steeping Duration t_{soak} (min)